

Cryoprecipitate for the treatment of life-threatening hemorrhage in children

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Abstract

Background: Hypofibrinogenemia is an important risk factor for poor outcomes in children with severe bleeding. There is a paucity of data on the impact of cryoprecipitate transfusion on outcomes in pediatric patients with life-threatening hemorrhage (LTH).

Study Design and Methods: This secondary analysis of a multicenter prospective observational study of children with LTH investigated subjects who were categorized by receipt of cryoprecipitate during their resuscitation and according to the etiology of their bleeding: trauma, operative, and medical. Bivariate analysis was performed to identify variables associated with 6-h, 24-h, and 28-day mortality. Cox Hazard regression models were generated to adjust for potential confounders.

Results: Cryoprecipitate was transfused to 33.9% (152/449) of children during LTH. The median (Interquartile range) time to cryoprecipitate administration was 108 (47–212) minutes. Children in the cryoprecipitate group were younger, more often female, with higher BMI and pre-LTH PRISM score and lower platelet counts. After adjusting for PRISM score, bleeding etiology, age, sex, RBC volume, platelet volume, antifibrinolytic use and cardiac arrest, cryoprecipitate administration was independently associated with lower 6-h mortality, Hazard Ratio (95% CI), 0.41 (0.19–0.89), ($p = 0.02$) and 24-h mortality, Hazard Ratio (95% CI), 0.46 (0.24–0.89), ($p = 0.02$).

Conclusion: Cryoprecipitate transfusion to children with LTH was associated with reduced early mortality. A prospective randomized trial is needed to determine if cryoprecipitate can improve outcomes in children with LTH.

KEYWORDS

cryoprecipitate, mortality, pediatrics, transfusion, trauma

1 | INTRODUCTION

Life-threatening hemorrhage (LTH) in children may occur from multiple etiologies, including trauma,

perioperative bleeding, gastrointestinal bleeding, and disseminated intravascular coagulation (DIC). Trauma is the most common cause of death in children >1 year of age, according to the Centers for Disease Control and

Abbreviations: DCR, damage control resuscitation; DIC, disseminated intravascular coagulation; GCS, glasgow coma scale; LTH, life-threatening hemorrhage; MTP, massive transfusion protocol; PRISM, pediatric risk of mortality III; RBC, red blood cell.

Prevention.¹ Death from traumatic hemorrhagic shock typically occurs early, within the first 6 h of injury,² and factors associated with death and massive transfusion in this population include shock and coagulopathy.^{3–7} The Massive Transfusion in Children (MATIC) study, a multicenter prospective observational study of approximately 450 pediatric patients with LTH, reported that while trauma was the most common etiology for life-threatening bleeding in the cohort (46%), a comparable proportion of children had surgical bleeding (34%) and medical illnesses (20%).⁸ Twenty-eight-day mortality was 37.5% and higher among children with medical bleeding (65.2%) compared with trauma (36.1%) and operative (23.8%).

To prevent death from hemorrhagic shock, Damage Control Resuscitation (DCR) transfusion strategies, based on military and adult civilian trauma data, are used to rapidly identify and treat traumatic hemorrhagic shock.⁹ A central tenant of DCR is hemostatic resuscitation which advocates for avoidance of hemodilution, transfusion with either whole blood or RBCs, plasma, and platelets in a 1:1:1 unit ratio, and the appropriate use of high fibrinogen-containing products (fibrinogen concentrates or cryoprecipitate).⁹ The use of DCR for non-traumatic etiologies of LTH is common despite a lack of high quality data to support its use in both children and adults.^{10,11}

Fibrinogen plays a critical role in hemostasis as a contributor to platelet aggregation, clot formation, and fibrinolysis.¹² Hypofibrinogenemia is an important risk factor for bleeding in clinical settings, including pediatric surgery.¹³ Cryoprecipitate and fibrinogen concentrate both effectively restore fibrinogen levels.¹⁴ In injured adolescents receiving massive transfusion, use of cryoprecipitate was associated with reduced 24-h mortality in the overall cohort and reduced 7-day mortality in those with penetrating trauma receiving ultra-massive transfusion (greater than 100 mL/kg of total blood products). While cryoprecipitate and fibrinogen concentrate may be given to critically ill children in the setting of surgery, trauma, and medical illnesses, there is a paucity of data on the impact of cryoprecipitate transfusion on outcomes in children with LTH.

In this secondary analysis, we hypothesize that children receiving cryoprecipitate in the setting of LTH will have improved mortality at 24 h compared to children who did not receive cryoprecipitate.

2 | METHODS

This is a hypothesis-generating, post hoc secondary analysis of the MATIC study. The methods of this study have previously been published.⁸ In brief, this was a prospective observational study of children with life-threatening

bleeding events. Children 0–17 years old were eligible for enrollment if they received greater than 40 mL per kg of total blood products over 6 h or if they were transfused under MTP activation. Data extracted from hospital medical records included demographics, cause of bleeding, Glasgow Coma Scale (GCS), vital signs, measures of end-organ function, fluids, blood products, hemostatic adjuncts, morbidities, and mortality. The Injury Severity Score (ISS) was collected from patients with traumatic injury and Pediatric Risk of Mortality III (PRISM) scores were calculated as previously described.⁸ For the analyses, children were categorized into receipt of cryoprecipitate versus no receipt of cryoprecipitate, and according to the etiology of their bleeding: trauma, operative, and medical.

2.1 | Analysis

Data were summarized as count and percentage or median (interquartile range). Transfusion volumes that were given during the LTH event were recorded and standardized by subject weight (mL/kg). The main outcome was mortality 24 h after the onset of the life-threatening bleeding event. This primary outcome is based on consensus recommendations from an international group of experts convened by the NIH and US Department of Defense.¹⁵ Secondary outcomes included 6-h and 28-day mortality as well as in-hospital complications (acute respiratory distress syndrome, sepsis, and acute kidney injury). Bivariate analyses with odds ratios were performed to identify variables associated with 6-h, 24-h, and 28-day mortality. Variables with no more than 10% of values missing were included in the bivariate analyses. Cox Hazard regression models were then generated for mortality at these same timepoints considering all variables that differed between groups on bivariate analysis (age, sex, race, BMI, PRISM score, bleeding etiology, tachycardia, volume of blood products administered, use of hemostatic adjuncts and use of Factor VIIa), as well as covariates associated with mortality (ethnicity, temperature, hypotension, crystalloid volume, and cardiac arrest) using a significance threshold of $p < 0.2$. Complete cases were included. A backward stepwise covariate selection technique was used to generate a parsimonious model and avoid overfitting. Covariates with evidence of collinearity with other potential confounders defined as variance inflation factor (VIF) > 10 were not included in the model. Adjusted survival curves were then generated based on Cox Hazard regression models. All hypothesis tests were two-sided and a p value < 0.05 was considered significant. Hazard Ratios (HR) are reported as HR, (95% Confidence Interval). Statistical analyses were conducted using Stata Statistical Software, version 16 (StataCorp, College Station TX).

3 | RESULTS

A total of 449 children with life-threatening bleeding met inclusion criteria (cohort demographics and characteristics published previously).⁸ In total, 152 (33.9%) children received cryoprecipitate during the LTH event. Children in the cryoprecipitate group were younger, more often female, with higher BMI and pre-LTH PRISM score and

lower platelet counts (Table 1). In the group that received cryoprecipitate the etiology of LTH was most often due to surgical bleeding (surgical 47%, trauma 32%, medical 21%), whereas in the group that did not receive cryoprecipitate, the trauma cohort was most common (surgical 28%, trauma 53%, medical 19%; $p < 0.001$). The most common types of surgery in the operative group were cardiothoracic (34%), neurologic (24%), and general (19%).

TABLE 1 Comparison of cryoprecipitate and no-cryoprecipitate groups.

	No cryoprecipitate (<i>n</i> = 297)	Cryoprecipitate (<i>n</i> = 152)	<i>p</i> value
Demographics			
Age (years) (<i>n</i> = 449)	8 (2–15)	5 (0.5–14)	0.007
Sex (<i>n</i> = 449)			
Male	172 (58)	75 (49)	0.09
Female	125 (42)	77 (51)	
Race (<i>n</i> = 380)			
White	146 (60)	96 (71)	<0.001
Black	88 (36)	21 (16)	
Other	11 (4)	18 (13)	
Ethnicity: Hispanic/Latino (<i>n</i> = 392)	26 (10)	17 (13)	0.50
Body mass index (<i>n</i> = 321)	19 (16–23)	17 (14–23)	0.003
Blood group (<i>n</i> = 453)			
A	98 (34)	59 (39)	0.54
B	40 (14)	25 (16)	
AB	8 (3)	4 (3)	
O	139 (49)	63 (42)	
Indication for transfusion (<i>n</i> = 449)			
Trauma	158 (53)	49 (32)	<0.001
Operative	82 (28)	71 (47)	
Medical	57 (19)	32 (21)	
Pre- life threatening hemorrhage (LTH) variables			
PRISM score (<i>n</i> = 428)	12 (5–21)	15 (8–24)	0.004
Lowest Glasgow Coma Scale score (<i>n</i> = 280)	8 (3–15)	6 (3–14)	0.31
Lowest temperature (<i>n</i> = 339)	36.2 (35.4–36.6)	36.1 (35.1–36.6)	0.54
Age-adjusted hypotension (<i>n</i> = 443)	163 (55)	92 (62)	0.22
Age-adjusted tachycardia (<i>n</i> = 422)	164 (59)	99 (68)	0.08
Hematocrit (<i>n</i> = 420)	29.4 (23.4–35.5)	28.9 (24–34.5)	0.9
International normalized ratio (INR) (<i>n</i> = 398)	1.4 (1.2–1.7)	1.4 (1.2–2.0)	0.16
Platelet count (<i>n</i> = 333)	180 (85–265)	121 (59–217)	0.003
Base deficit/excess (<i>n</i> = 389)	0 (–5 to 6)	0 (–5 to 5)	0.66
Fibrinogen before LTH	182 (142–246)	212 (114–293)	0.45
	(<i>n</i> = 183)	(<i>n</i> = 112)	
Fibrinogen after LTH	194 (157–251)	234 (189–309)	0.003
	(<i>n</i> = 154)	(<i>n</i> = 93)	

During the LTH, children who received cryoprecipitate were transfused a greater total volume of blood product at 120 mL/kg (67–236) compared to children who did not at 41 mL/kg (20–77) ($p < 0.001$) as well as greater volumes of individual component products (Table 2).

The median (QR) time to cryoprecipitate administration was 108 (47–212) minutes. Medical indications had the longest time to cryoprecipitate administration at 153 (50–285) minutes, compared to 130 (71–212) minutes for trauma and 74 (15–175) minutes for surgical indication ($p = 0.005$). The median (IQR) fibrinogen concentration before the initiation of the resuscitation for LTH was 212 (114–293) mg/dL in the cryoprecipitate group ($n = 112$) compared to 182 (142–246) mg/dL in the no cryoprecipitate group ($n = 183$), ($p = 0.45$). The fibrinogen concentration post LTH was 234 (189–309) mg/dL in the cryoprecipitate group ($n = 93$) versus 194 (157–251) mg/dL in the no cryoprecipitate group ($n = 154$), ($p = 0.003$).

On univariate analysis, unadjusted mortality at 6 h was 10% in the cryoprecipitate group compared to 18% in the no cryoprecipitate group ($p = 0.03$); at 24 h it was 18% in the cryoprecipitate group versus 24% in the no cryoprecipitate group, ($p = 0.23$); and at 28 days mortality was 43% in the cryoprecipitate group compared to 35% in the no cryoprecipitate group, ($p = 0.12$). Covariates associated with mortality included age, ethnicity, PRISM score, bleeding etiology, temperature, hypotension, platelet volume, antifibrinolytics, crystalloid volume, and cardiac arrest (Table S1). After Cox Hazard regression model generation controlling for PRISM score, bleeding etiology, age, sex, volume RBCs, volume platelets, antifibrinolytic use and cardiac arrest, cryoprecipitate administration was independently associated with lower 6-h mortality, HR (95% CI), 0.41 (0.19–0.89), ($p = 0.02$) and 24-h mortality, HR (95% CI), 0.46 (0.24–0.89), ($p = 0.02$) (Table 3). Adjusted survival curves

TABLE 2 Blood product administration and unadjusted mortality compared between cryoprecipitate group and no cryoprecipitate group.

	No cryoprecipitate ($n = 297$)	Cryoprecipitate ($n = 152$)	<i>p</i> value
Blood products and adjuncts			
Duration of massive transfusion event hrs ($n = 449$)	2.7 (1.0–5.0)	4.8 (3.0–7.7)	<0.001
Total volume mL/kg ($n = 449$)	41 (20–77)	120 (67–236)	<0.001
RBC volume mL/kg ($n = 449$)	24 (12–46)	56 (28–121)	<0.001
Plasma volume mL/kg ($n = 449$)	12 (0–24)	32 (16–67)	<0.001
Platelet volume mL/kg ($n = 449$)	3 (0–11)	21 (10–42)	<0.001
Cryoprecipitate volume mL/kg ($n = 449$)	-	4 (2–9)	-
Plasma deficit ($n = 449$)	10 (0–24)	19 (2–59)	<0.001
Platelet deficit ($n = 449$)	18 (8–35)	31 (10–83)	<0.001
Plasma:RBC ratio ($n = 449$)	0.5 (0–0.85)	0.6 (0.36–0.91)	0.002
Platelet:RBC ratio ($n = 423$)	0.13 (0–0.32)	0.31 (0.18–0.56)	<0.001
Antifibrinolytics (yes/no) ($n = 449$)	31 (10)	25 (16)	0.07
Factor VII (yes/no) ($n = 447$)	17 (6)	39 (26)	<0.001
Crystalloid volume mL/kg ($n = 445$)	18 (1–50)	14 (1–57)	0.58
Outcome			
Mortality – 6 h ($n = 69$)	54/297 (18)	15/152 (10)	0.03
Death due to bleeding	41/69 (76)	13/15 (87)	0.49
Death during LTH	13/54 (24)	7/15 (46)	0.11
Mortality – 24 h ($n = 99$)	71/297 (24)	28/152 (18)	0.23
Mortality – 28 days ($n = 168$)	104/297 (35)	64/152 (43)	0.12
Acute respiratory distress syndrome	52 (17)	29 (26)	0.05
Sepsis	31 (10)	13 (9)	0.62
Acute kidney injury	38 (13)	45 (30)	<0.001

TABLE 3 Cox hazard regression models for 6 and 24-h mortality.

	6-h mortality			24-h mortality		
	Hazard ratio	95% confidence interval	p value	Hazard ratio	95% confidence interval	p value
Cryoprecipitate (yes/no)	0.41	0.21–0.79	0.008	0.48	0.28–0.80	0.006
PRISM score	1.04	1.02–1.06	<0.001	1.05	1.03–1.07	<0.001
Indication						
Trauma (ref)						
Operative	0.32	0.15–0.70	0.004	0.38	0.21–0.69	0.002
Medical	1.27	0.72–2.23	0.40	1.10	0.68–1.78	0.71
Age (years)	0.95	0.91–0.99	0.02	0.96	0.92–0.99	0.02
Female sex	1.31	0.79–2.18	0.30	1.28	0.83–1.96	0.26
Volume red blood cells mL/kg	1.01	1.00–1.01	0.02	1.00	1.00–1.00	0.02
Volume platelet mL/kg	0.98	0.96–1.00	0.11	0.99	0.98–1.01	0.44
Antifibrinolytic	0.34	0.10–1.09	0.07	0.47	0.22–1.03	0.06
Cardiac arrest	2.30	1.35–3.93	0.002	2.63	1.68–4.12	<0.001

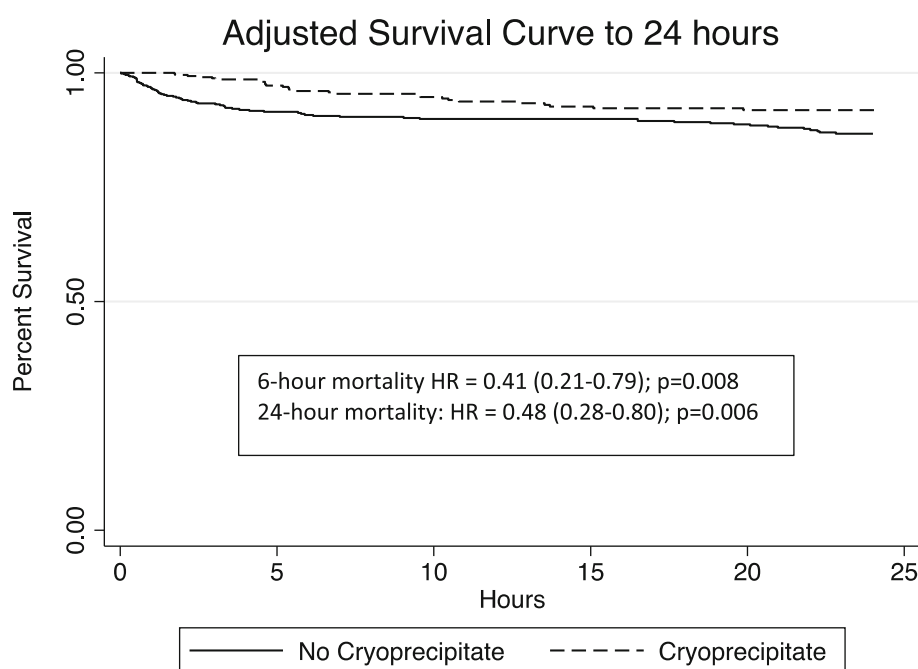


FIGURE 1 Adjusted survival curve up to 24 h after LTH.

depict this survival advantage at both 6 and 24 h (Figure 1).

4 | DISCUSSION

This hypothesis-generating, post hoc analysis of a prospective study assessed the association between the use of cryoprecipitate during the LTH on mortality in 449 children. This study is unique since we examined children with bleeding of diverse etiologies and included cryoprecipitate given

during the LTH event. These data suggest that after adjusting for multiple potential confounders, the use of cryoprecipitate in children during LTH is independently associated with improved survival at 6 and 24-h, but not at 28 days.

It is common, but not universal, for pediatric trauma centers to use cryoprecipitate for the resuscitation of bleeding children. In a survey of 46 pediatric hospitals, 61% (28/46) of sites indicated that cryoprecipitate was included in their massive transfusion protocols (MTPs)¹⁰ compared to 49% (65 of 132) of adult trauma centers that incorporate cryoprecipitate into MTP

policies.¹¹ Increased incorporation of cryoprecipitate into MTPs may expedite its use with the potential to improve outcomes in patients, if our findings are replicated in randomized controlled trials.

Fibrinogen is essential for hemostasis. Fibrinogen deficiency in severely bleeding patients may be due to consumption and/or reduced production. Hypofibrinogenemia has been reported during hyperfibrinolysis, which can occur in trauma, post cardiopulmonary bypass, with extracorporeal membrane oxygenation, and DIC.¹⁶⁻¹⁹ Cryoprecipitate may be used to replenish fibrinogen, in addition to factor VIII, XIII, and vWF, which are also important for hemostasis. Cryoprecipitate may be manufactured from a single whole blood donation, which can then be pooled in to varying volume sizes, or produced from a single plasma apheresis unit.²⁰

In this study, the pre-LTH values for fibrinogen were close to the lower limit of the normal range (200 mg/dL) before the severe bleeding started. The pre-LTH values between the two study groups are also not statistically different indicating that the groups were starting off at a similar baseline for fibrinogen concentration. The post-LTH fibrinogen values did not increase much from the pre-LTH values, potentially because the amount of fibrinogen that was replaced during the resuscitation with fibrinogen containing products (cryoprecipitate, plasma, and platelet units) was similar to the amount of fibrinogen consumed during the LTH. It is difficult to know if the increased post-LTH differences in fibrinogen in the cryoprecipitate group compared to the non-cryoprecipitate group are related to our findings of an independent association with the use of cryoprecipitate with increased 6 and 24-h survival. The other coagulation factors in cryoprecipitate such as vWF, Factors VIII and XIII may have also contributed to our findings.

In the adult literature, cryoprecipitate transfusion in combination with plasma²¹ and tranexamic acid²² was associated with increased survival compared to blood components and hemostatic adjuncts alone. However, little is known about optimal use of cryoprecipitate in pediatric life-threatening hemorrhage. Tama et al. examined a retrospective cohort of injured children with the mean age of 16 years receiving massive transfusion. In this cohort, those receiving cryoprecipitate had a significantly lower 24-h mortality when compared with those who did not and a significantly lower 7-day mortality in children with penetrating trauma and those transfused at least 100 mL/kg of total blood products.²³ The median age of children in our study was 5 years in contrast to 16 years in the study by Tama. Our data suggest that cryoprecipitate may be advantageous for the treatment of LTH in younger children, who are known to have a unique and

developing hemostatic system at baseline compared to adolescents and adults. There are two small pilot RCTs that compared cryoprecipitate to fibrinogen concentrates in young children requiring cardiac surgery. Both trials suggest feasibility and safety for performing larger definitive trials with no differences in outcomes.^{14,24}

Cryoprecipitate was most commonly transfused to patients with LTH from operative bleeding, and was given most expeditiously from the onset of LTH in this population. It is possible that the increased and earlier use of cryoprecipitate contributed to the improved adjusted outcomes at 6 and 24 h for children with operative LTH. Other reasons for the improved outcomes could include relatively normal baseline physiology and hemodynamics of surgical patients prior to the LTH compared to trauma patients who may have prolonged shock and coagulopathy prior to treatment of LTH.

No differences in 28-day mortality were noted between study groups, suggesting that other disease processes or therapeutic interventions may have contributed to the late mortality results. The increased acute respiratory distress syndrome and acute kidney injury on unadjusted analyses in the cryoprecipitate treated group might suggest that there are potential thrombotic complications, though survivor bias may have contributed to these findings since there were more patients alive after 24 h in the cryoprecipitate treated group. A recent National Heart Lung and Blood Institute and the United States Department of Defense sponsored workshop, comprised of international leaders in trauma trials and care, recommended that 24-h mortality was the optimal outcome for effectiveness in trials of hemostatic agents for severe bleeding, whereas 28-day mortality was more appropriate as a safety outcome.¹⁵ While this is controversial, the FDA has agreed with using early mortality outcomes (6- or 24-h mortality) in recently developed trauma trials. Importantly, the improved early survival may allow for interventions to address other events in the interim to ultimately address late survival.

4.1 | Limitations

Limitations to this study include the post hoc analysis of a prospective observational study database, with a high degree of missing data for some variables. The inclusion criteria selected for children with massive bleeding, limiting generalizability to children with non-massive bleeding. Few centers reported TEG data, precluding an analysis of global functional measures of hemostasis that may explain the survival benefit of cryoprecipitate or why cryoprecipitate was given to patients with higher pre-LTH fibrinogen concentrations. Fibrinogen concentrates were not used at

any centers that participated in this study, therefore any association between this treatment and outcomes could not be reported. The analysis combines children with multiple etiologies of LTH; the number of children in each subgroup (trauma, operative, medical) did not permit a robust analysis within each of these cohorts.

4.2 | Conclusions

In conclusion, this study suggests that the use of cryoprecipitate is independently associated with improved outcomes in children with LTH and that use earlier in a child's resuscitation may be most beneficial. Prospective randomized trials are needed to assess if cryoprecipitate can improve survival after LTH. Optimally, platform trials should be performed to allow for examining multiple hemostatic resuscitation therapies at once to determine which combination of therapies are both effective and safe in children with life-threatening hemorrhage.

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CONFLICT OF INTEREST STATEMENT

Philip C. Spinella: Scientific Advisory Board for Cerus. Cassandra D. Josephson: Medtronic, LLC; Octapharma LLC; Cellphire, LLC.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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